PROJECT OVERVIEW

Safety and accessibility,

connected at the wrist.

BINDU

Standalone Safety and Accessibility Wearable

DEVELOPMENT STAGE

Pre-hardware integration - firmware scaffold and two parts ordered

The problem

Older adults living independently may struggle to request help after a fall or emergency. Blind and low-vision users need accessible obstacle feedback, while caregivers need to know when a safety wearable has been removed.

TARGET USERS

Older adults living independently, blind and low-vision adults, and their family caregivers. Each user can enable the functions relevant to their needs.

Abstract

Bindu is a standalone 4G wrist-wearable project combining physical SOS, motion-based fall detection, watch-removal alerts and gesture-activated obstacle awareness. A forward-facing distance sensor will drive vibration pulses that become faster as obstacles approach, with optional audible feedback.

An ESP32-S3 will coordinate local sensing and send remote events over Wi-Fi or LTE data to a self-hosted backend. The backend will request caregiver SMS through httpSMS and a dedicated Android gateway phone. The wearer will not need a nearby phone; remote delivery still depends on internet and gateway availability.

Development proceeds from subsystem tests to an integrated PCB and custom enclosure. Evaluation will measure gesture reliability, false alarms, ranging limits, alert delivery and power use. Location, vitals and Telugu/Hindi voice interaction are planned extensions. No detection or battery-performance results are claimed at this stage.

PRODUCT AND USER EXPERIENCE

One device. Clear interactions.

Four planned functions connect accessible local feedback with caregiver awareness.

01 Physical SOS

A deliberate button press creates an emergency event. The watch uploads it over an available data connection; the backend records the event and requests caregiver SMS. A cancel input and clear local feedback are part of the design.

02 Gesture-activated obstacle mode

Raise the forearm so the sensor faces forward, then perform the deliberate wrist gesture. Distinct vibration confirms activation. Faster pulses indicate a nearer obstacle; optional sound adds a close-obstacle warning. Repeat a validated gesture to deactivate.

03 Watch-removal awareness

Confirm sustained contact loss before alerting the caregiver. Filter loose-strap events, provide an approved charging/removal interval, and report stable worn-again contact. Missing heart-rate readings alone are not proof of removal.

04 Motion-based fall detection

Replace fixed sensor stubs with real IMU readings, then evaluate impact and stillness logic on controlled recordings. Tune false alarms and detection thresholds before enabling automatic caregiver alerts.

KEY ENGINEERING DISTINCTION

A watch-mounted IMU measures watch motion. Bending only the hand may leave the forearm and watch stationary. The requested gesture needs experiments and may require an additional wrist-bend sensor.

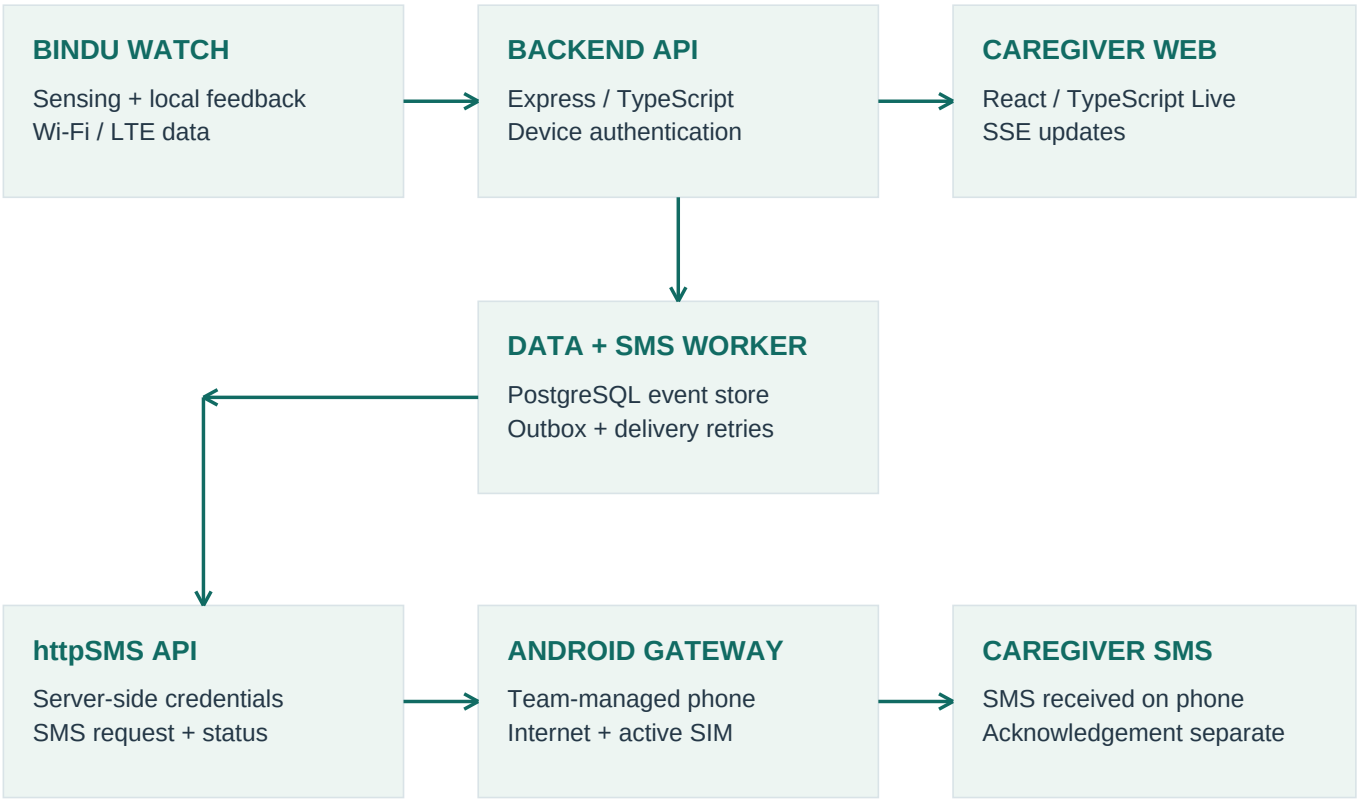
WEARABLE DESIGN DIRECTION

Use the compact XIAO controller, an integrated carrier PCB, selected LTE module, protected flat battery and a custom wrist enclosure. Sensor direction, antenna clearance, optical contact and battery current capability must be checked together. Case dimensions and runtime are not yet validated.

SYSTEM ARCHITECTURE

Local awareness. Remote alerts.

Selected architecture - software implementation and Kubernetes deployment are pending.



The gateway phone can stay with the team; it is not paired to the wearer.

Layer	Selected technology
Firmware	C, ESP-IDF, FreeRTOS; I2C, UART, cellular AT commands
Frontend	React, TypeScript, Vite; responsive caregiver dashboard
Backend / data	Node.js, TypeScript, Express, PostgreSQL; HTTPS + SSE
Deployment	Container images; existing Kubernetes VM; persistent storage
Notifications	httpSMS REST API and dedicated Android gateway phone

Open-source application stack; no new managed hosting subscription is required. Existing VM/storage, mobile data, carrier SMS and fabrication costs remain separate.

READINESS AND DELIVERY

What exists. What comes next.

A measurable delivery plan with explicit prototype status.

AVAILABLE IN THE REPOSITORY

ESP-IDF/FreeRTOS task scaffold, event dispatch, priorities, power-mode state and component boundaries. Hardware inputs, SMS, voice and sync remain stubs; selected-board build is unverified.

HARDWARE ORDERED

XIAO ESP32-S3 Plus from HUBTRONICS; MPU6050 from Quartz. Receipt is not yet confirmed. Other sensors, cellular integration and the custom PCB/enclosure are pending.

Delivery gates and acceptance evidence

Gate	Required evidence
1. SOS to caregiver	Wi-Fi disabled; LTE upload reaches backend and gateway SMS arrives. Measure latency and record failure/recovery.
2. Local assistance	Measure obstacle range limits, gesture false triggers and warning recognition in supervised tests.
3. Removal + motion	Test loose straps, charging, restored contact and controlled motion; record false alarms and missed events.
4. Wrist integration	Verify PCB fit, supply stability, antenna performance and measured battery runtime before final case assembly.

OPERATING BOUNDARIES

Obstacle warnings supplement a cane or guide dog; a quiet sensor does not prove a clear path or detect every drop-off. Remote alerts require watch data, backend and Android gateway availability. httpSMS lists a free quota of up to 200 sent/received SMS per month; carrier allowances still apply. Persist alert jobs, deduplicate retries and distinguish API acceptance, delivery and caregiver acknowledgement.

PROOF OF WORK AND TECHNICAL REFERENCES

Project repository httpSMS API Plans Kubernetes